



Meteorological drought severity forecasting utilizing blended modelling ☆,☆☆



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ABSTRACT

Prediction of droughts has recently become imperative as frequency and intensity are increasing mostly due to climatic variation. Indeed, drought is a highly significant disaster that results in widespread damage to all kinds of ecosystems, agricultural production systems, and water resources systems. Accurate techniques of forecasting are necessary for the purpose. Conventional methods lack the intricate time-space correlation in meteorological data. The research proposes an ensemble of Extreme Gradient Boosting (XGBoost), Long Short Term Memory (LSTM), and Tabular Network (TabNet) for a higher accuracy in drought forecasting. With the large meteorological dataset that involves temperature, precipitation, humidity, and wind speed as features, the model integrates:

- The tree capabilities of XGBoost perform feature selection very effectively.
- Temporal Pattern Analysis using LSTM.
- Insight obtained from the attention mechanism-based TabNet.

Empirical results demonstrate that the proposed ensemble outperforms individual models, achieving the lowest Root Mean Square Error (RMSE: 0.6582) and Mean Absolute Error (MAE: 0.5377), and the highest Coefficient of Determination (R^2 : 0.5069). Furthermore, it yields the best Nash–Sutcliffe Efficiency (NSE: 0.5107) and Kling–Gupta Efficiency (KGE: 0.6039), confirming its superiority in drought severity forecasting. The ensemble outperforms traditional models, aiding early drought warnings and water conservation planning.

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Background

The impetus behind the approach is improvement of the precision of drought prediction in view of the tightening grip that climate change imposes on drought events, which have become increasingly more severe and frequent. The traditional methods of forecasting drought often fail because they are not capable of accounting for complex interactions in time and space between meteorological data [1]. These traditional methods, usually based on past data and linear models, cannot capture the nonlinear relationships and interdependencies of many variables, which is an important aspect of capturing drought dynamics [2,3]. The study in [4] focuses on environmental and economic effects of drought by providing a detailed overview of models of Deep Neural Networks (DNN). The evaluation shows how dataset analysis is complicated and utilized to forecast drought using Convolutional Neural Networks, Multi-Layer Perceptron (MLP), and hybrid models. However, it faces difficulty in recognizing certain constraints, such as spatial resolution, data quality, and it struggles in implementing it in real-time. The utilization of Artificial Neural Networks (ANNs) model for drought indices prediction, like Standardized Precipitation Index (SPI) and the Effective Drought Index, is the key component of the study proposed in [5]. The SPI and remote sensing data utilized to analyze conditions of drought [6]. They examines rainfall data from 1990 to 2014 and LANDSAT ETM+ satellite imagery to assess the drought severity across Kutch, Gujarat, a region that experiences severe water shortages. To enhance the accuracy of drought prediction system, they integrates the meteorological data and satellite images to address the requirement for continuous drought monitoring. However, the study's reliance on satellite-derived rainfall data may introduce spatial and accuracy limitations. In Ethiopia's Awash River Basin, study [7] tested how well Machine Learning (ML) predicts short-term droughts using SPI. Among the models—Artificial Neural Networks (ANN), Support Vector Regression (SVR), and a hybrid Wavelet-ANN—the WA-ANN gave the most accurate results for 1- and 3-month forecasts. However, it didn't cover long-term drought patterns. Authors in [8] explored how well hybrid ML models can predict meteorological drought in Türkiye. They compared models using drought indices—SPI and SPEI—and measured accuracy using MAE, RMSE, R^2 , and NSE. They found that prediction errors actually decreased as the timescale increased. However, due data unavailability, others can't easily verify or reproduce the results. The study in [9] uses the Standardized Streamflow Index (SSI) as input data to develop a model based on a Deep Belief Network. The approach is compared with models such as SVR and the MLP. The proposed model demonstrates the excellent accuracy and less prediction RMSE and MAE. However, the study is limited to exclusive use of SSI dataset which is relatively small that can restrict full potential of model. Authors in [10] examine the performance of several ML algorithms, and DNN, for predicting drought in various climatic conditions of the Rajasthan desert region. Because of variations in quality of data and climatic behavior, it successfully enhances short-term drought prediction but finds challenging in long-term predictions.

Method details

This section describes the proposed methodology. The methodology takes advantage of an ensemble approach that combines XGBoost, LSTM, and TabNet, each with strengths that help address these complexities in meteorological data. XGBoost's decision tree-based structure is especially strong at identifying relationships between features, which is very important for a good feature selection process. The architecture of the LSTM model allows the network to capture temporal dependencies in the data, making the model very suitable for sequential data such as weather patterns. Meanwhile, the attention mechanism of TabNet enhances the interpretability of the model by giving it the ability to concentrate on the most influential features, which is very crucial in understanding which variables impact drought severity the most. The ensemble approach is integrated motivated by the need for a more robust, adaptive, and interpretable model that improves the prediction accuracy but more importantly gives a better understanding of the factors influencing drought. The resulting models are then into serving as a useful tool in the planning of water conservation as well as early warning systems for resource management and mitigation of risk.

The methodology followed sequences:

1. Collecting Meteorological Data
2. Cleaning and preprocessing the data
3. Training individual models (XGBoost, LSTM, TabNet)
4. Combining predictions using ensemble model
5. Validation process
6. Analyzing and interpreting the model's performance

The structured approach of the method integrates the diverse strengths of various models to yield accurate robust predictions of drought, thus providing proactive strategies to manage and conserve water.

The process is composed of the following steps:

Data collection

The Dataset used in the analysis of the study contains the meteorological data related to the drought conditions. The dataset includes 50,000 rows, which includes the key indicators of the weather for predicting the drought conditions based on the weather factors. This helps to identify the specific patterns in the dataset for the prediction purposes. The Data includes parameter levels ranging from 0 to 4 with zero being the lowest and five being the highest to identify the severity of the drought. The features, which are included in the dataset, are related to the weather such as temperature, dew point, precipitation, wind speed, humidity, drought severity and various other measurements related to meteorological. The study focuses on the past historical data of multiple years as

Table 1
Dataset description.

Temperature (°F)	Precipitation (inches)	Humidity (%)	Wind Speed (mph)	Drought Severity Index	Region	State	Drought Duration (days)
87	0.12	45	12	2.5	Northeast	NY	5
90	0	38	10	2.8	Northeast	NY	6
88	0.15	50	11	2.1	Northeast	NY	7
85	0.08	42	13	2.3	Southwest	TX	3
92	0.02	40	15	3	Southwest	TX	4
91	0.1	47	16	2.9	Southeast	FL	8
89	0.05	43	14	2.4	Southeast	FL	9
86	0.18	52	11	2.2	Midwest	IL	2

input to the modelling. The dataset has been used to identify the drought conditions of the region based on the weather factors and based on the complex relationships between the meteorological factors and the drought severity, it enables to have a detailed study and based on that have a model that would identify and forecast future occurrences of drought conditions. The Weather data serves as the input for the Modelling. The [Table 1](#) shows the dataset description.

Preprocessing

To analyze and work with the dataset it is very important that as to make sure that the dataset which is being used has to be preprocessed for the model building in order to get optimal performance. Data preprocessing plays a vital role in ensuring efficient model analysis. To pre-process the raw dataset, non-relevant columns that were redundant and did not improve the predictiveness of the target variable score were deleted. The helps in removing noisy and irrelevant attributes so that the more robust model is being deployed. KNN Imputation was used to impute missing values. The method, without estimating and filling in missing values with similarity between data points, avoids bias and noises. Since mean and median imputation were too simplistic generalizations of complex patterns involved in the data, they had to be avoided. Since there was no over-powering of any one feature during the model training, standardization was used so that the features were kept uniform. All the feature variables were scaled with zero mean and unit variance for that dataset. The applied dimension reduction to avoid overfitting and efficient computation is Principal Component Analysis (PCA). Considering the variance retention criteria, number of components were reduced to 10 principal components to ensure balancing between explained variance and dimensionality. The resulted in a cumulative explained variance of over 95 %, thus ensuring that most of the variation in the data was retained. An 80:20 ratio split the dataset into training and testing sets.

Conceptual design

Drought severity prediction in this work was conducted using a supervised machine learning framework, with the drought severity index (0 to 4) as the output variable. The model described here as shown in [Fig. 1](#), integrates an ensemble of XGBoost, Long Short-Term Memory (LSTM), and TabNet to tap their strengths. XGBoost for succinct feature selection, LSTM for learning temporal dependency, and TabNet for attention-based interpretability on tabular data. The methodology involved collecting and preprocessing meteorological variables like temperature, precipitation, humidity, and wind speed. Preprocessing steps included KNN imputation for missing values, normalization using standard scaling, and dimensionality reduction by preserving significant variance through Principal Component Analysis (PCA). Both models were separately trained and tuned using hyperparameter tuning and their predictions stacked using a stacking ensemble method where XGBoost was used as the meta-learner to generate the final drought severity predictions.

The features involved in the dataset used for the study are the meteorological features gathered for drought prediction. The score variable ranges from 0 to 4, means drought levels, and is used as the target variable. During the preprocessing step, columns that appear less relevant to the feature set are removed. Standard Scaler used for normalization of the feature set, which ensures that every feature is on equal footing to contribute to the model. Following PCA application to reduce the dimensionality and select only the most relevant components, ten components were retained. Additionally, the PCA is vital in enhancing the computational efficiency and preventing overfitting by selection of fewer components that catch the greatest variance as evident from the ratio of cumulative explained variance by the selected components.

Because of their respective benefits, XGBoost, LSTM, and TabNet were used for the individual models. The Optuna was used for the hyperparameter optimization package in order to optimize the decision tree-based robust ensemble approach of XGBoost. The most significant hyperparameters, which include `n_estimators`, `learning_rate`, `max_depth`, and `subsample`, are fine-tuned and selected to achieve the optimal model performance. The significance of each parameter owes its control of the complexity and learning rate of the model to a fair trade-off between bias and variance. In the setup of the LSTM model, two layers of 50 units each were used. These can handle time-related dependencies very well and Dropout layers to avoid overfitting. Following the LSTM layers, there was dropout, which de-activated neurons randomly during training at a rate of 0.2 in order to avoid overfitting. Regularization is fundamental in order to assure better generalization and to make the model not dependent on some particular patterns in the data.

The dimensionality of the decision prediction layer is set by using the parameter `n_d`, which has been set at 51. It further influences the ability of the model to detect complex patterns. The model will learn associations that are more complex in case the value of

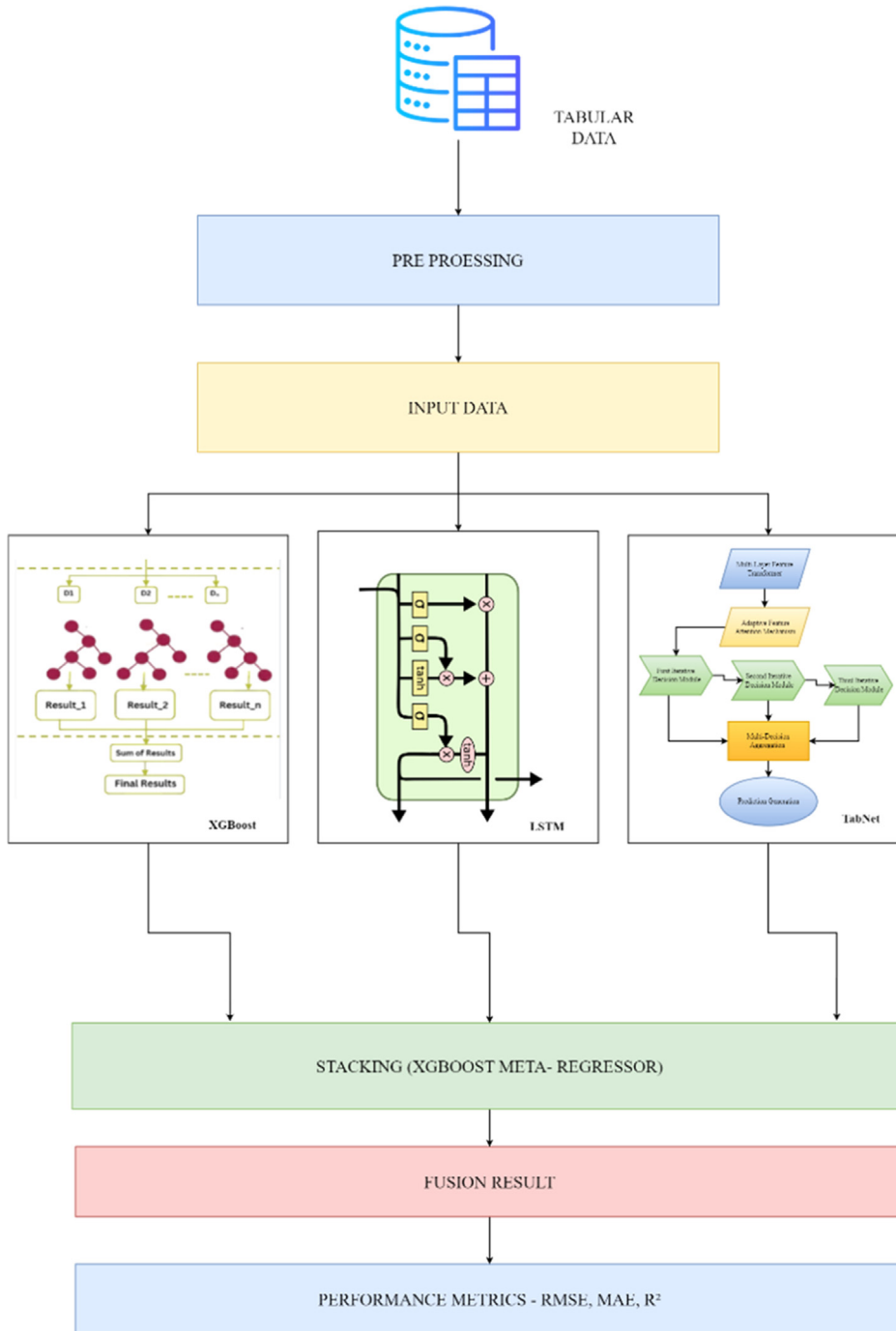


Fig. 1. Proposed system.

n_d is increased; the computational requirement increases with that. The number of dimensions in the attention mechanism, which decides on where to focus, is set to 38 with n_a . The lower the value of n_a is, the less likely the model is overfitting, and hence it is more interpretable and easier. With the value n_{steps} , which determines the number of steps or iterations of decision in TabNet, a model can improve sequentially, thereby refining its predictions. Defaults to 3. By being set to 1.16, TabNet balances exploration and exploitation through gamma in training. Ensuring the model chooses only the most critical characteristics for each decision step while encouraging feature sparsity, a sparse regularisation (λ_{sparse}) was set to 0.0008565797 to increase interpretability and reduce overfitting. Sparsemax mask type had been selected for feature selection due to the fact that it produces sparse attention

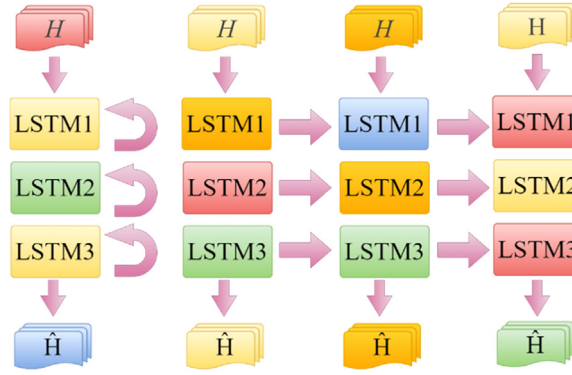


Fig. 2. Stacked LSTM model architecture.

weights representing the most important features compared to softmax mask and thus increases the interpretability of a feature. The Adam optimizer set to a learning rate of 0.00342 did further refinement of convergence.

Finally moving on to development of stacking ensemble model, which pools together the advantages of the different models. The combination approach chosen is meta-model stacking, which trains a meta-model using XGBoost and includes the predictions of LSTM, TabNet, and XGBoost all as input. Since stacking allows leveraging the strength of various algorithms and enhances prediction accuracy through the learning process as one builds on their strengths, it was preferred over other forms of hybrid strategy, such as bagging or boosting. The underlying rationale is that stacking covers more significant strengths of each learner in contrast with boosting, which selectively corrects the errors for weaker learners stepwise. In stacking, there is the meta-learner that may adaptively assign weights to the models, which increases generalization; it is different from bagging, which averages the predictions. Finally, the models and the stacking ensemble were tested using R2 Score, MAE, RMSE. There has been a marked improvement in RMSE, and the stacking model always outperformed the separate models, which indicated that combining the predictions of XGBoost, LSTM, and TabNet through stacking was capturing complementing patterns that were not caught by each individual model.

The improved performance accounts for the ensemble's ability to combine the feature selection of structured decision trees, the sequence learning capability of LSTMs, and the adaptive feature learning for tabular data of TabNet.

Stacked LSTM model

The Stacked LSTM is a part of RNN which creates a DNN by combining multiple layers of LSTM stacked on top of each other. When the output of one LSTM layer becomes the input of the next LSTM layer, it becomes the stacked LSTM architecture. Due to this, the long-term dependencies are captured and result in learning more complex representations of the input sequence. It can be modelled owing to its high capacity due to more parameters as compared to that of LSTM and thus, can perform better. Each layer learns different features when stacked LSTM exists. These kinds of features allow the combination forming a more meaningful representation of the data. That is why Stacked LSTM performs feature representation about data more. The architecture of Stacked LSTM is shown in Fig. 2.

As depicted in Fig. 2, the input is passed to the first LSTM layer. This first LSTM layer then processes the sequence and generates the output. Then this output is fed as an input to the second layer of LSTM. The second layer processes the output to generate the output. This process is repeated for every additional LSTM layer in the stack. Horizontal arrows denote the flow of information between LSTM units across time steps to capture temporal dependencies. The output from the LSTM layer has to be mapped to the required output dimension-for example, prediction of the next value in a time series or classification for an input sequence given. Output from the final LSTM layer is fed into a fully connected output layer. The fully connected output layer is a group of neurons, having learnable weights and biases; it uses the activation function like ReLu for generating the final output of the network. The Stacked LSTM model consists of 4 layers of LSTM. These layers are based on the hyperparameter tuning concept which showed the best performance for 4 layers of LSTM, epochs of the model set to 25 and the batch size set to 10. These layers will provide better understanding of the data. Training a model with 25 epochs can help improve its performance and allow it to capture more complex patterns and trends. This increases the efficiency of the prediction. The LSTM Layer configuration is shown in Eq. (1) - (15).

LSTM layer 1,

$$\text{Input gate} : i_t = \sigma(w_{i1} * [h_t - 1, x_t] + b_{i1}) \quad (1)$$

$$\text{Forget gate} : f_t = \sigma(w_{f1} * [h_t - 1, x_t] + b_{f1}) \quad (2)$$

$$\text{Output gate} : o_t = \sigma(w_{o1} * [h_t - 1, x_t] + b_{o1}) \quad (3)$$

$$\text{Cell gate} : ct_1 = f_{t1}ct - 1 + i_{t1} \tanh(w_{c1} * [h_t - 1, x_t] + b_{c1}) \quad (4)$$

$$\text{Hidden layer : } h_{t1} = o_{t1} * \tanh(c_{t1}) \quad (5)$$

LSTM layer 2,

$$\text{Input gate : } it2 = \sigma(wi2ht1 + ui2ht1 + bi2) \quad (6)$$

$$\text{Forget gate : } ft2 = \sigma(wf2ht1 + uf2ht1 + bf2) \quad (7)$$

$$\text{Output gate : } ot2 = \sigma(wo2ht1 + uo2ht1 + bo2) \quad (8)$$

$$\text{Cell gate : } ct2 = ft2ct1 + it2tanh(wc2ht1 + uc2ht1 + bc2) \quad (9)$$

$$\text{Hidden layer : } h_{t2} = o_{t2} * \tanh(c_{t2}) \quad (10)$$

LSTM layer 3,

$$\text{Input gate : } it3 = \sigma(wi3ht2 + ui3ht2 + bi3) \quad (11)$$

$$\text{Forget gate : } ft3 = \sigma(wf3ht2 + uf3ht2 + bf3) \quad (12)$$

$$\text{Output gate : } ot3 = \sigma(wo3ht2 + uo3ht2 + bo3) \quad (13)$$

$$\text{Cell gate : } ct3 = ft3ct2 + it3tanh(wc3ht2 + uc3ht2 + bc3) \quad (14)$$

$$\text{Hidden gate : } h_{t3} = o_{t3} * \tanh(c_{t3}) \quad (15)$$

Where,

i_t is the input gate activation vector at time t

f_t is the forget gate activation vector at time t

o_t is the output gate activation vector at time t c_t is the cell gate activation vector at time t h_t is the hidden state activation vector at time t

w, wc, uc are the weight matrices for the input, forget, output and cell state gates, and for the cell state update respectively

bc are the bias vectors for the input, forget, output and cell state gates, respectively x_t is the input vector at time t

σ is the sigmoid activation function

$\tanh$ is the hyperbolic tangent activation function

XGBoost model

The XGBoost approach is employed in the proposed work. In order to minimize anticipated errors, XGBoost trains a set of Decision Trees (DT) iteratively and optimizes their weights. The hyperparameters of XGBoost model are selected carefully for the application to ensure better performance. The XGBoost model architecture is depicted as shown in the Fig. 3.

The XGBoost model architecture is based on the ensemble of DT. One by one each DT analyzes input data and comes with a result that is based on prediction. Such individual prediction gets added and it gives final output of that model. XGBoost models get the final outcome by taking the cumulative summation of each and every one's result from a tree that builds one after other. It's a kind of learning method to boost the resultant output accuracy which will be possible when multiple weak models are employed.

TabNet model

TabNet is an attention-based deep learning model that processes tabular data. Because of the special architecture, combining the interpretability of decision trees with the strength of neural networks, it can be utilized for both high-performance predictions and the analysis of feature importance. The method of TabNet selectively focuses on a subset of features at each decision step using a sequential attention technique. The dynamic feature selection process helps TabNet gain sparsity and improve generalization based on the most relevant features for each data point. Among its key contributions is that the prime technique it introduces is sparse attention strategy sparsemax whereby the model is forced to focus on only the most important attributes for every sample. It enhances interpretability through the removal of unnecessary complexity, thereby resulting in improved generalization performance.

The above Fig. 4 depicts the architecture of TabNet model. It shows that the architecture starts with an input layer which takes a tabular data source in its raw form. The model is receiving meteorological data, which includes temperature, humidity, precipitation, wind speed, and so on. The Feature Transformer Block uses a number of nonlinear layers in order to transform raw features from input. The high-level representations of the data more suitable for the further stages of the decision-making procedure are extracted from the

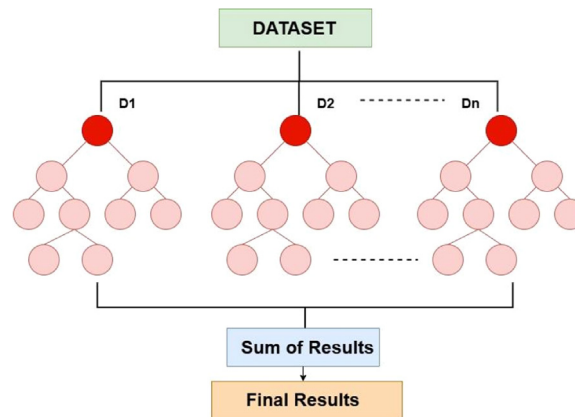


Fig. 3. XGBoost model architecture.

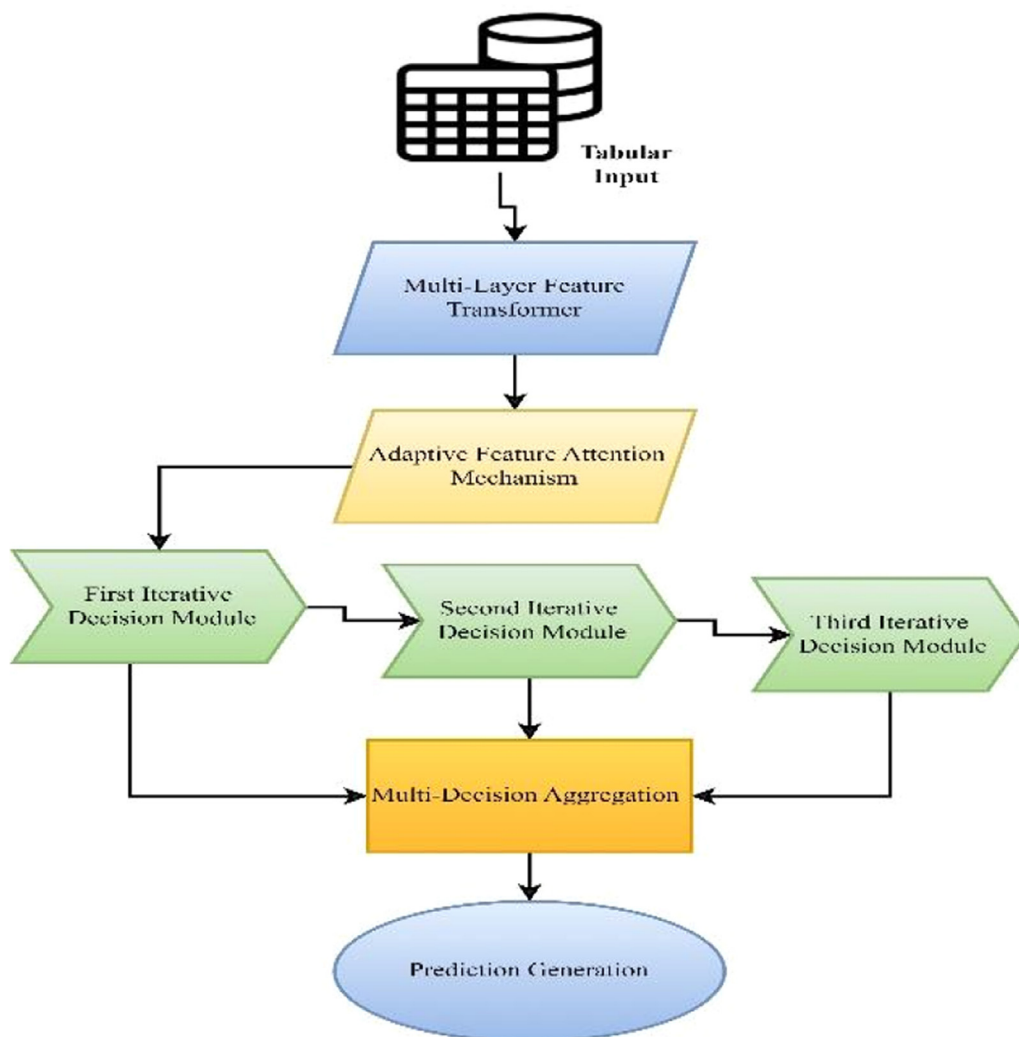


Fig. 4. TabNet architecture.

Table 2
Hyperparameter tuning table.

Parameter	Value
XGBoost	
_n_estimators	500
_learning_rate	0.05
_max_depth	6
_min_child_weight	3
_subsample	0.8
_colsample_bytree	0.8
LSTM	
_units_layer1	50
_dropout_layer1	0.2
_units_layer2	50
_dropout_layer2	0.2
_epochs	30
_batch_size	32
_optimizer	adam
_loss	mean_squared_error
TabNet	
_n_d	51
_n_a	38
_n_steps	3
_gamma	1.16
_lambda_sparse	0.0008565797
_mask_type	sparsemax
_optimizer	Adam
_learning_rate	0.00342
_batch_size	1024
_virtual_batch_size	128
_max_epochs	200
_patience	30
Stacking Parameters	
Stacking_meta_model	
Stacking_meta_model_n_estimators	100
Stacking_meta_model_learning_rate	0.1

incoming data. The Attentive Transformer Block, utilizing a sparse attention mechanism is followed below. The model can selectively focus on different feature subsets for each decision step with the help of the technique. At each step, TabNet makes a dynamic decision about which features are most important. By ensuring that only a subset of features is considered at each decision step, sparse attention reduces overfitting and improves the interpretation of the model. A feature transformer and an attentive transformer block are incorporated into each of the next sequential judgement phases. As these steps of the decision-making process are sequenced, the means that the outcome of one step leads into the next, gradually enhancing the prediction. The Aggregation Layer aggregates the outcomes of each decision step. The layer generates the ultimate prediction with the help of all the knowledge it learns at every step of the stage. The aggregation process makes sure that the final decision of the model is made on the basis of multiple points of view the data has, each of these being in turn affected by the considerations of a number of feature subsets at the decision steps. The final forecast output of the model is derived through the combining of all the data with its ultimate forecasting on the Forecast Layer.

Proposed system

The proposed system intends to forecast drought severity by using the meteorological variables temperature, precipitation, humidity, and wind speed with XGBoost, LSTM, and TabNet models. Each of the models implemented its strength: XGBoost utilized the optimization of hyperparameter tuning to optimize decision tree-based learning, LSTM identified the time dependency, and TabNet used selective attention for important features. A stacking ensemble strategy is used to combine the models, which led to an increase in the accuracy of prediction to utilize the power of each model. Data pre-processing involved scaling, imputation, and a reduction in dimensionality using PCA; R2 Score, MAE, and RMSE were the basis of model evaluation. The ensemble performs better than the individual models with a more robust approach toward drought severity prediction.

The Table 2 has the hyperparameters used in the XGBoost, LSTM, and TabNet model and the meta-model. The important hyperparameters for XGBoost include n_estimators, learning_rate, max_depth, min_child_weight, subsample, and colsample_bytree. These parameters give the number of boosting rounds, learning rate, depth of trees, and proportion of samples and features used at training to provide a measure of success in the model.

The parameters for the table of the LSTM model are numbers of units in each of the LSTM layers, dropout rates to prevent overfitting, and adam as the optimizer used in training. Similarly, the parameters for the table of TabNet include n_d and n_a controlling the decision prediction and attention layers, respectively, and n_steps, gamma, lambda_sparse, and mask_type, which control the complexity of the model as well as sparsity. Finally, the table makes emphasis on the stacking meta-model based on XGBoost by

Table 3
Result evaluation.

Model	RMSE	MAE	R2
XGboost	0.6591	0.5400	0.5055
LSTM	0.7443	0.6060	0.3694
Tabnet	0.8344	0.6740	0.2074
Ensembled Model	0.6582	0.5377	0.5069
[13] XGBoost and LSTM and [14] ensemble model	0.845	0.667	0.845

showing tailored parameters and the usage of all above models in order to exploit different strengths to improve overall predictive performance.

Method validation

The developed methodology validated for predicting drought events using a set of quantitative performance metrics: predictive accuracy and robustness of the ensemble model. The kind of objective benchmarking, based on its performance in the given framework, gives the model credibility for implementing it in practice. For the purpose of the study, the three evaluation metrics were chosen to be RMSE, MAE, and R^2 Score. These parameters proves potential to be used as reliable mechanism of prediction by adaptability with time-dependent shifting environmental conditions. Also NSE, KGE parameters have been evaluated to perform comprehensive assessment of how well a model replicates observed drought patterns over time [11,12]. RMSE calculates the square root of the average of the squared differences between observed and predicted values. Therefore, it indicates the difference in the extent to which the model's predictions resemble actual data. A lesser RMSE value will point out a better fit for predicting drought severity with high accuracy. MAE quantifies the average absolute size of the errors made without any regard to the direction in which the prediction lies, making the model results more straightforwardly interpretable. The R^2 Score measures the proportion of variance in the observed data that the model can explain. The greater the R^2 score, the better the model's ability to capture important patterns in the data, validating the overall strength of its prediction. The performance compared for the ensemble model against that of separate models such as XGBoost, LSTM, and TabNet. The overall trend was to be coherent since the RMSE/MAE should be at the lowest end, whereas R^2 will be a value greater than 1 since it is based on 20 different independent models to find the ensemble model.

XGBoost learns from its decision trees; LSTM recognizes temporal dependability; TabNet has its attention mechanisms. Consequently, it all sums up for better predictions. Such ensemble superiority there is an advantage in using a combination model - in the case, temporal, structural, as well as feature-focused insights are effectively balanced. To ensure further, generalized performance metric measurements of the developed model and therefore more generalized assessment of ensemble capabilities, cross-validation was involved with its validation across different subsets of the complete dataset applied to confirm consistency and reliability of the ensemble approach.

As shown in the Table 3, the Ensembled model, performed quite well with RMSE of 0.6582, MAE of 0.537 and R^2 of 0.5069. The XGBoost model also demonstrated a great performance with RMSE of 0.6591, MAE of 0.54 and R^2 of 0.505. While comparing with a study by Wu et al. [13] and Singh et al. in [14], which utilized neural network and statistical architectures for predicting drought severity, as shown in the Table 3, the proposed ensemble and Individual model demonstrates better performance. Specifically, the ensemble model in the study by Wu et al. [13] and Singh et al. [14] produced an RMSE of 0.845, MAE of 0.667, and R^2 of 0.845.

Finally, the proposed study successfully integrates the XGBoost, LSTM, and TabNet models to predict drought severities using a stacking ensemble technique. As such, the ensemble optimizes the decision tree-based learning that accompanies XGBoost, the temporal dependencies of LSTMs, and the most important aspects of TabNet. The results on the accuracy of the ensemble were highly improved in forecasting drought severities since the pooled models exploited characteristics particular to each individual technique. It had critical preprocessing involved in the steps of scaling, imputation, and dimensionality reduction through PCA, as they led to models being trained on cleaned and relevant data, which performed very well and eradicated all the chances of overfitting.

The Taylor diagram in above Fig. 5 was used to compare the correlation, standard deviation, and root mean square error between the observed drought severity values and predicted values for the standalone models—XGBoost, LSTM, and TabNet—and the new ensemble. The diagram graphically displayed that the ensemble model correlated most with the observed values and, therefore, its better predictive power and stability among the individual models.

The network graph in Fig. 6 displays the performance correlations among the individual models—XGBoost, LSTM, TabNet—and the ensemble (Stacking) model over evaluation metrics such as RMSE, MAE, and R^2 . The width of the connecting edges represents the strength of association, such that the wider the edge, the stronger the performance linkage or better agreement with the concerned metric. Observably, the ensemble (Stacking) model has robust correlations with both MAE and RMSE, which attest to its higher accuracy and consistency over the individual models. The ensemble model demonstrated better performance than individual models using the benchmarking criteria of R^2 Score, MAE, and RMSE also the NSE and KGE. Table 4 indicates the NSE and KGE for all the models.

Importantly, it delivered a more robust framework for prediction. Fine-tuning the hyperparameters of the important parameters finally resulted in optimal performance for each model—just refer to the hyperparameter table. The stacking ensemble model that makes use of XGBoost as a meta-model. In the way, it would take full advantage of the predictive strengths of LSTM and TabNet

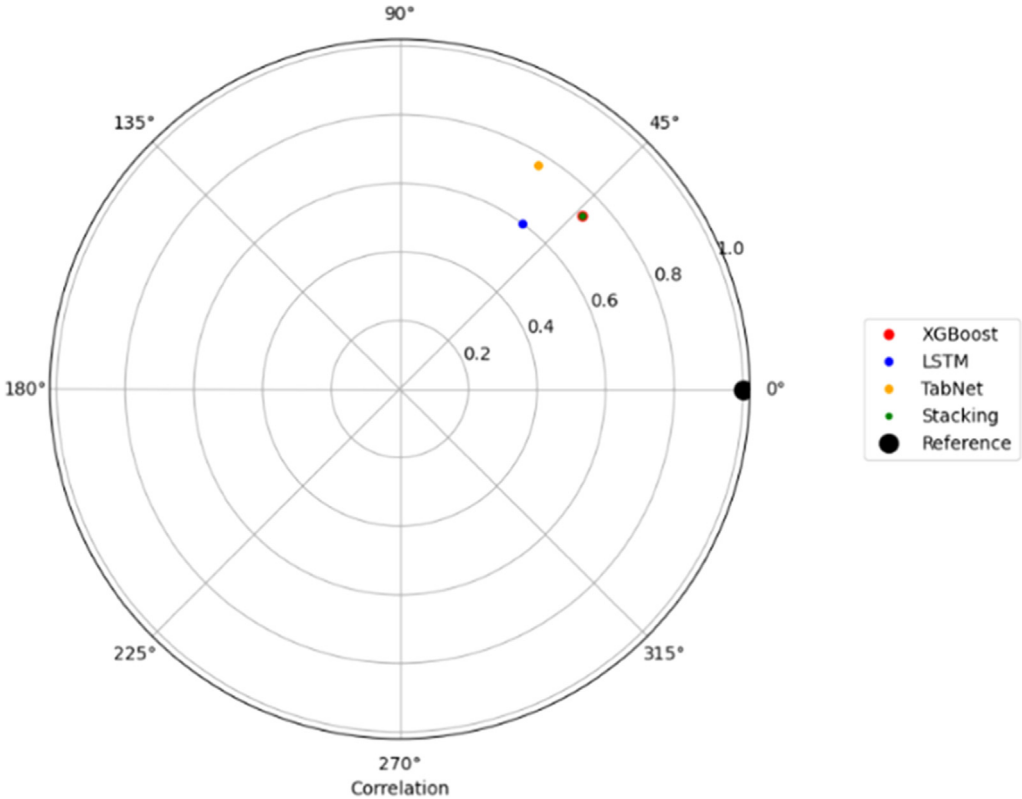


Fig. 5. Taylor diagram.

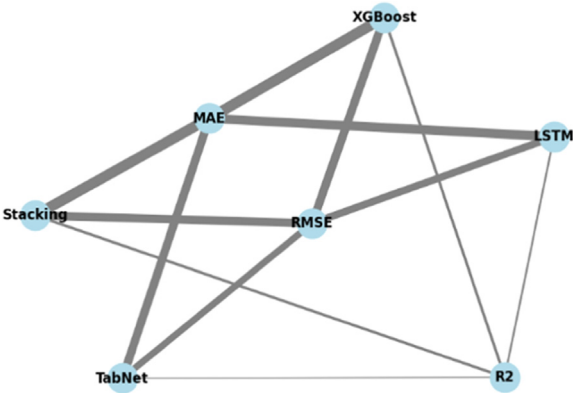


Fig. 6. VOS-model performance network.

Table 4
Model evaluation metrics.

Model	NSE	KGE
XGBoost	0.5083	0.6025
LSTM	0.3532	0.4317
TabNet	0.2166	0.4694
Ensemble Model	0.5107	0.6039

together, leading to an accurate generalizable system in predicting drought severity. It is the way, which provides a comprehensive and successful mechanism dealing with the issues of forecasting drought conditions.

In the work of drought severity forecasting the study has contributed majorly utilizing deep learning and transfer learning methodologies finally showcasing high potential for real life application. The study also puts into light that scalability of the method for handling complicated dataset and tackling the drought effects.

Limitations

The proposed modeling approach registers promising outputs for predicting the severity of meteorological droughts, but there are certain limitations. The historical meteorological datasets are utilized and while these span a broad range, they might be poorly representative of sudden and extreme changes in climate now becoming increasingly prevalent over recent years. Since the model is validated on rigorous datasets, wider deployment to various climatic regimes and comparison to a broader list of algorithms might form the basis of further research. In short, while the proposed ensemble model demonstrates high accuracy in meteorological drought forecasting, it is limited by its reliance on historical data and untested generalization across diverse climatic regions. Future work will focus on integrating real-time meteorological streams, expanding to multi-regional datasets, and enhancing the model's applicability for drought early-warning systems and water resource planning.

Ethics statements

No animal experiments were performed, nor were animal subjects used in the research work presented in this paper. The dataset is collected from <https://www.kaggle.com/datasets/cdminix/us-drought-meteorological-> data.

CRedit author statement

Aaditya Ahire: Writing – original draft, Writing – review & editing, Investigation, Validation. **Nilima Zade:** Writing – original draft, formal analysis, project administration, resources. **Umeed Mujawar:** Writing – Methodology, Writing – original draft, Conceptualization. **Dimple Mehata:** Writing – Methodology, Writing – original draft, Conceptualization. **Ketan Kotecha:** Writing – Methodology, Writing – original draft, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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